Predictive framework for Customer Attrition

NLP and Supervised Statistical Learning models for BCG X

Federico Baio

May 2025

1 Project overview

This project focused on developing a robust churn prediction framework for a phone company by integrating structured customer profiles with unstructured complaint data. By merging records for 7,043 clients with 1,605 unique complaints, we identified high-risk segments that would have remained invisible using simpler analysis.

2 System architecture & Processing logic

The following pseudocode details the pipeline used to unify qualitative and quantitative signals before feeding the predictive engine:

```
1.  PREPROCESS D_clients (7043 records):
    a. Remove 11 NAs in TotalCharges [cite: 331]
    b. Resolve collinearity (TotalCharges vs. Tenure * MonthlyCharges):
    a. Tokenize text and remove stop-words
    b. Extract sentiment polarity via Bing Lexicon
    c. Compute WeightedScore based on complaint intensity
3.  FUSION: Inner-join datasets via CustomerID
4.  VALIDATION:
    a. Initialize 5-Fold Cross-Validation (Seed 123)
    b. Grid search for Gradient Boosting hyperparameters
5.  SELECTION: Deploy Lasso for optimal feature sparsity
```

Listing 1: Algorithm: Multi-Source Data Fusion and Prediction

3 Machine learning strategy

3.1 Model benchmarking and validation

We implemented a rigorous 5-fold cross-validation (CV) to guarantees model stability and avoid overfitting. Four primary architectures were used:

- **Lasso regression:** Implemented with an L1 penalty ($\alpha = 1$), choosen also to perform automated feature selection.
- **Logistic regression:** The simplest one, served as the baseline for linear separability.
- **Gradient boosting (GBM):** Subjected to extensive hyperparameter tuning including tree depth (1–3), shrinkage factors (0.01–0.1), and iteration counts (300–700).
- **Random forest:** Configured with 300 trees to evaluate non-linear bagging performance

3.2 Performance metrics

The Lasso model achieved the lowest misclassification error (18.46%). Interestingly, tree-based models (Random Forest and Boosting) did not yield significant improvements over linear methods. This indicates a predominantly linear decision boundary within the feature space, where the L1 regularization of Lasso effectively handled the high dimensionality introduced by the NLP metrics.

Table 1: Model performance evaluation (5-Fold CV)

Model architecture	**Misclassification error**
Lasso regression	**18.463%**
Logistic regression	18.578%
Gradient boosting (GBM)	18.703%
Random forest	18.928%

4 Strategic findings

- **The efficiency of sparsity:** Lasso successfully identified and brought to zero redundant predictors, such as the collinear relationship between tenure and total charges.
- **Sentiment significance:** In the final models, the engineered *WeightedScore* and the number of complaints were among the most statistically significant predictors of imminent churn.
- **High-Value attrition:** Fiber optic users exhibit a 32% churn rate, compared to 7% for those without internet. This highlights a specific service-level fragility in the premium segment.
- **Retention drivers:** Two-year contracts and tenure are confirmed as the primary anchors of customer stability[cite: 252].